

JUAN MADHY

AI Engineer · Full Stack Developer · AIoT Specialist

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+62 888-0538-5353 · Indonesia (Open to Remote)

PROFESSIONAL SUMMARY

Independent engineer specializing in AI automation, full-stack web development, and AIoT systems. Proven track record of delivering field-ready solutions — from deploying AI detection systems on rural livestock farms to building scalable ERP platforms and industrial IoT tools. Currently deepening expertise in Agentic AI and LLM-powered workflows. Open to remote opportunities at the intersection of cutting-edge AI and real-world engineering impact.

TECHNICAL SKILLS

AI & Automation	AI Automation · Agentic AI · LLM Integration · AI Workflow Design · AI Service Integration
Full Stack Web	React.js · Next.js · Node.js · Express.js · JavaScript · TailwindCSS · HTML · CSS
Database	PostgreSQL · MySQL
IoT & Embedded	ESP32 · ESP8266 · C/C++ · Modbus RTU/TCP · MQTT · HTTP API Integration · Sensor & Actuator Control
DevOps & Tools	Docker · Git/GitHub · Linux CLI · Figma · VS Code

EXPERIENCE

- AIoT Engineer Intern** · [Chickin Indonesia](#)

Jul – Dec 2024 · Indonesia

 - Deployed IoT and AIoT solutions across 4 cities: Yogyakarta, Solo, Boyolali, and Bandung.
 - Designed and built a portable Modbus sensor configuration tool — enabling field engineers to label, troubleshoot, and reset sensor addresses on-site, eliminating office dependency for sensor configuration.
 - Served as Person-in-Charge (PIC) for AI-based animal detection deployment across partner rural livestock farms.
 - Led hardware assembly, firmware flashing, QA/QC testing, and client-site installation of AIoT products.
- Founder & IoT Developer** · [SIF Skaneka \(Smart Integrated Farming\)](#)

Jun 2023 – Present · Indonesia

 - Founded and led end-to-end development of an integrated smart farming product — product design, market analysis, SWOT, hardware engineering, and full-stack software implementation.
 - Built an automated fish feeding and plant watering system with aquaponic water recycling, enabling resource-efficient and sustainable farming.
 - Awarded Silver Medal at FIKSI 2023 (National Innovation & Entrepreneurship Festival) organized by PUSPRESNAS, Indonesia's National Achievement Center.

SELECTED PROJECTS

- TFE ERP** · Full Stack Developer

2025 – Present

Stack: React.js · Express.js · PostgreSQL · Docker · TailwindCSS

Comprehensive ERP platform with task dashboards, progress tracking, team assignment, and meeting notes — scalable from small teams to enterprise operations. erp.juan.web.id
- SmartHome AI** · AI & IoT Engineer

2025 – Present

Stack: ESP32-S3 · C++ · Node.js · React.js · AI Integration

Complete AIoT smart home ecosystem featuring a live AI assistant for real-time intelligent device control and natural language home management.
- Nevada — Smart Hydroponics** · IoT Developer

2025

Stack: ESP32 · C++ · Node.js · React.js · PostgreSQL · TailwindCSS

Full-stack hydroponics monitoring system with real-time sensor dashboards, automated nutrient management, and IoT device integration. nevada.juan.web.id
- Portable Sensor Tool** · Embedded Engineer

2024

Stack: ESP32 · C/C++ · Modbus RTU · LCD UI

Industrial-grade portable tool built for Chickin Indonesia field engineers to configure, label, and troubleshoot Modbus sensors on-site. Eliminated office dependency for sensor management.

EDUCATION

Business Information Systems · State Polytechnic of Malang	2025 – Present
Computer Network & Telecommunications Engineering · SMKN 1 Kadamangan	2022 – 2025

CERTIFICATIONS & ACHIEVEMENTS

Achievements

Silver Medal · National Innovation & Entrepreneurship Festival (FIKSI 2023) — PUSPRESNAS	2023
Top 10 National · SOBAT Competitions, Technology Category — United Tractors	2024

Courses & Certifications

Industrial IoT: Modbus RTU/TCP with MQTT & Node-RED — PT Avisha Inovasi Indonesia	2024
ERP & Artificial Intelligence Future Technology Workshop — Maspion IT Surabaya (Microsoft)	2024
IoT Intern Program — Chickin Indonesia	2024
Expo Inovasi Vokasi (INOVAKS) — Malang State Polytechnic	2025
Ardumeka Podcast Participant — PT Avisha Inovasi Indonesia	2025